INSTITUTO
COLLEGE

MATH 110: General Education Math

Fall 2026

Homework 4

Name: _____

1. (*6 pts*) At a clinic, 40% of the patients smoke, 25% have high blood pressure, and 15% smoke and have high blood pressure. A patient is chosen at random. Let S be the event "the patient smokes" and H the event "the patient has high blood pressure." Find $P(H \mid S)$ and $P(S \mid H)$, and describe each one in words.

2. (*6 pts*) Two fair dice are rolled. Let A be the event "the numbers the dice land on add up to 8" and B the event "both dice land on even numbers." Write A by listing its sample points, and find $P(B \mid A)$.

3. (*6 pts*) At a hospital, 60% of the patients are admitted through the emergency department, and 15% of these patients are admitted to the intensive care unit (ICU). Of the patients who are not admitted through the emergency department, 5% are admitted to the ICU. A patient is chosen at random. Let E be the event "the patient is admitted through the emergency department" and I the event "the patient is admitted to the ICU." Find $P(E \cap I)$, $P(E^c \cap I)$, and $P(I)$.

4. (*6 pts*) At a blood bank, 9% of the donors have type O-negative blood. Six unrelated donors give blood on a certain morning. What is the probability that exactly 1 of them has type O-negative blood?

5. (*6 pts*) Let A and B be events with $P(A) = 0.3$ and $P(B) = 0.5$.

(a) Suppose A and B are mutually exclusive. Find $P(A \cap B)$ and $P(A \cup B)$.

(b) Suppose instead that A and B are independent. Find $P(A \cap B)$ and $P(A \cup B)$.

6. (*14 pts*) A clinic follows 400 of its patients through one flu season. The table shows how many of them are vaccinated against the flu and how many of them get the flu.

	Gets the flu	Does not get the flu	Total
Vaccinated	12	228	240
Not vaccinated	24	136	160
Total	36	364	400

One of the 400 patients is chosen at random. Let V be the event "the patient is vaccinated" and F the event "the patient gets the flu."

(a) (*3 pts*) Find $P(V)$, $P(F)$, and $P(V \cap F)$.

(b) (*3 pts*) Find $P(F \mid V)$ and $P(F \mid V^{\mathrm{c}})$.

(c) (*2 pts*) How many times as likely is an unvaccinated patient to get the flu as a vaccinated patient?

(d) (*3 pts*) Find $P(V \mid F)$, and describe it in words. Why is it different from $P(F \mid V)$?

(e) (*3 pts*) Are V and F independent? Explain.

7. (*14 pts*) A clinic classifies its patients by age group and by whether they have a primary care doctor. The table gives some of the probabilities for a randomly chosen patient.

	Has a primary care doctor	Does not have one	Total
Under 40	0.12		0.30
40–64		0.20	0.40
65 or older	0.21		
Total			1

Let Y be the event "the patient is under 40," O the event "the patient is 65 or older," and D the event "the patient has a primary care doctor."

(a) (*4 pts*) Fill in the missing probabilities in the table.

(b) (*3 pts*) Find $P(D \mid Y)$ and $P(D \mid O)$.

(c) (*3 pts*) Find $P(O \mid D)$, and describe it in words.

(d) (*2 pts*) Are O and D independent? Explain.

(e) (*2 pts*) Find $P(Y \cup D)$.

8. (*14 pts*) A fair coin is tossed 3 times. Let A be the event "the first toss lands on heads," B the event "exactly two of the tosses land on heads," and C the event "all three tosses land the same way."

(a) (*3 pts*) Write the sample space by listing its sample points, and write A, B, and C by listing their sample points.

(b) (*3 pts*) Find $P(B)$ and $P(B \mid A)$. Are A and B independent? Explain.

(c) (*3 pts*) Find $P(C)$ and $P(C \mid A)$. Are A and C independent? Explain.

(d) (*2 pts*) Find $P(A \mid B)$.

(e) (*3 pts*) Are B and C mutually exclusive? Are they independent? Explain.

9. (*14 pts*) A medicine cabinet holds 10 vials of a vaccine, and 3 of them are expired. A nurse checks the vials one at a time, in random order, until she finds an expired one. Let F_k be the event "the first expired vial is the kth vial she checks."

(a) (*2 pts*) Find $P(F_1)$.

(b) (*3 pts*) Use the General Multiplication Rule to find $P(F_2)$.

(c) (*3 pts*) Find $P(F_3)$.

(d) (*3 pts*) What is the probability that the nurse has to check more than 3 vials?

(e) (*3 pts*) Explain why $P(F_1) + P(F_2) + P(F_3)$ plus your answer to part (d) must equal 1, and check that it does.

10. (*14 pts*) At a clinic, 15% of the patients miss their scheduled appointments. A doctor has 8 appointments with unrelated patients on a certain morning, and we count how many of the patients miss their appointments. Let E_k be the event "exactly k of the patients miss their appointments."

(a) (*2 pts*) Explain why this is a binomial experiment. What are n, a success, and p?

(b) (*2 pts*) Find $P(E_0)$.

(c) (*3 pts*) Find $P(E_2)$.

(d) (*4 pts*) What is the probability that at most 1 of the patients misses an appointment?

(e) (*3 pts*) What is the probability that at least 2 of the patients miss their appointments?